GROUP 3: CAMPUSBOOK

Week 2 Group Project Submission

GitHub repository	https://github.com/Gerber-W/SPM_Group_Project
Project Supervisor	Jabulile Mashibini
Client	CTU Training Solutions
Submitted	Monday 19 October 2026

Name	Role
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Chapter 1: Introduction

Background of the Study

Consultations give students time with a lecturer outside class to ask questions get feedback and work through problems Research shows this support is often under used and that the reasons are usually practical rather than academic (Griffin, 2014)

At CTU Training Solutions arranging a consultation is informal Students send an email a Teams or WhatsApp message or simply knock on the lecturer door during posted hours Each lecturer handles this differently and nobody has a clear view of who is available who is coming or which meetings took place

Other sectors have solved a similar problem with online booking in healthcare web based appointment systems have reduced no-shows and waiting times and saved staff time (Zhao, 2017) and automated reminders improve attendance even further (Guy, 2012) With 96% of South African households owning a mobile phone (Statistics South Africa, 2024) a booking system that works well on a phone can reach almost every student

Problem Statement

CTU Training Solutions has no single reliable way for students to book consultations with lecturers Requests are spread across email Teams WhatsApp and walk-ins so students wait outside offices messages go unanswered and meetings are missed or double-booked Lecturers cannot plan their time because they do not know who is coming or why the problem peaks before tests and assignment deadlines exactly when students need help most

It affects students who need academic support and lecturers who juggle requests from several channels Our group has experienced this first hand our lecturer reviewed the proposal and agreed it addresses a real need and a short student survey will show how widespread it is

Research Aim

To design develop and evaluate CampusBook a web based system that lets CTU students book change and cancel consultations and lets lecturers manage their availability so that fewer consultations are missed or left unplanned

Research Objectives

1. Analyse how consultations are arranged today through a survey of at least 20 students by 21 October 2026
2. Compare at least three existing scheduling systems on their features strengths and limitations by 19 October 2026
3. Design and develop a web system with all six must-have features working on a live database by 23 October 2026
4. Test the system so that every must-have test case passes a booking takes under 2 minutes and pages load within 3 seconds
5. Evaluate the system with at least 5 students and 2 lecturers with 80% or more rating it easy to use

Research Questions

1. How do students and lecturers at CTU arrange consultations now and what problems do they face?
2. What do existing scheduling systems offer and what do they lack for a campus setting?
3. How can a web system let students book consultations and lecturers manage their time?
4. Does the system meet its functional and non-functional requirements?
5. Do students and lecturers find the system easy to use and useful?

Proposed Software Solution

CampusBook is a responsive web application with two roles Lecturers publish their weekly consultation slots and see who has booked and why Students browse open slots book one with a short reason for the visit and get an email reminder before the meeting The six must have features are:

1. Login with separate student and lecturer roles
2. Lecturers publish weekly consultation slots
3. Students book change or cancel a slot with a reason for the visit
4. Email confirmation and a reminder before each booking
5. Booking history for students and lecturers
6. Mobile friendly design

Scope

In scope	Out of scope
One CTU campus	Other campuses or institutions
Student and lecturer roles	Admin reporting dashboards
Six must have features with email reminders	Video calls SMS and native mobile apps
Test data created by the team	Links to CTU live student systems

Significance of the Study

- Students can see when lecturers are free get confirmed bookings and stop wasting trips before assessments
- Lecturers control their consultation time and know in advance who is coming and why
- CTU Training Solutions gains a low-cost tool that supports student success and evidence on whether online booking works as well on campus as it does in healthcare

Chapter 2: Literature Review

Existing Research

Consultations are under used Griffin (2014) surveyed undergraduate students at a large American university and found that most of the factors that stop students attending office hours are outside the lecturer control The one exception was feedback: students who found it useful came back more often A summary of similar survey work found that 66% of more than 600 students had never attended office hours for a course and only 8% went more than once a month (Weimer, 2015) Both sources show the problem clearly but they focus on student attitudes rather than on how meetings are arranged and neither tests a tool that makes booking easier

Online booking works in healthcare Zhao (2017) reviewed 21 web based medical appointment systems Online booking cut no shows with one system reporting 2% for web bookings against 8% for phone bookings and it also shortened waiting times raised satisfaction and reduced staff workload The review warns that providers worry about losing control of their schedules a concern lecturers are likely to share Guy (2012) add that automated reminders increase attendance These studies come from healthcare where appointments are more formal than a quick consultation so the idea still needs testing in education

People use systems that are useful and easy The Technology Acceptance Model suggests that people adopt a system when they find it useful and easy to use (Davis, 1989) Nielsen (1994) usability heuristics turn this into practical design rules such as always showing a clear confirmation and stopping a student from booking a slot that is already taken

Existing or Similar Systems

Microsoft Bookings offers a booking page that syncs with Outlook sends email reminders and creates Teams meetings It comes with Microsoft 365 business and education plans but SMS reminders only work in North America and the UK and need an extra Teams Premium licence Microsoft It is built for businesses serving customers so it has no idea of students lecturers or modules and setting it up needs administrator access

Calendly is a popular polished scheduling tool Its free plan allows only one event type and one calendar and reminders need the Standard plan at 10 US dollars per lecturer per month Calendly Students would need a separate link for every lecturer and booking data would sit with a third party

Moodle Scheduler is a Moodle plugin that lets teachers create slots for students to book with reminders booking limits and attendance tracking (Moodle, 2017) It suits teaching well but it only works inside one course needs an administrator to install it and is only an option for institutions that run Moodle

	Microsoft Bookings	Calendly	Moodle Scheduler	CampusBook
Lecturer sets slots	Yes	Yes	Yes	Yes
Student books and cancels	Yes	Yes	Yes	Yes
Free reminders	Email only	No paid plan	Yes	Yes email
Student and lecturer roles	No	No	Course roles	Yes
All lecturers in one place	Yes	No	No per course	Yes
Extra cost	Microsoft 365 licence	Per lecturer	Needs Moodle	None
Main drawback	Business focus	Cost and data privacy	Tied to one course	Must be maintained

Lessons for CampusBook All three systems handle the basic act of booking well so CampusBook copies what works: clear open slots instant confirmation and reminders Where they fall short is fit Microsoft Bookings and Calendly are designed for businesses and Calendly charges per lecturer for reminders Moodle Scheduler fits teaching but stays inside a single course CTU current mix of email Teams WhatsApp and walk-ins costs nothing but offers no shared view of availability and no record of meetings

Technology Review

The chosen tools are among the most widely used in the 2025 Stack Overflow Developer Survey so documentation and help are easy to find (Stack Overflow, 2025)

CampusBook stores names email addresses and booking details which counts as personal information under the Protection of Personal Information Act (Republic of South Africa, 2013) The system will collect only what it needs hash all passwords show each user only their own bookings and use made up data during development

Part	Choice	Why	Alternative
Front end	React	Builds pages from reusable components Meta used by about 45% of developers	Plain HTML and JavaScript harder to keep consistent
Back end	Node.js and Express	One language JavaScript across the whole team used by about 49% of developers	ASP.NET which adds a second language
Database	MySQL	Relational tables for users slots and bookings and the team already knows it	PostgreSQL more popular at about 56% but less familiar to us
Reminders	Email	Free and reaches everyone	SMS which costs per message
Method	Scrum	Sprints of a month or less with a review after each (Schwaber and Sutherland, 2020) which fits our weekly deadlines	Waterfall too rigid for weekly feedback

Research Gap

Research shows that consultations help students but are under used (Griffin, 2014) and that online booking with reminders reduces missed appointments in healthcare (Guy, 2012; Zhao, 2017) Existing tools serve businesses or single Moodle courses well but none gives CTU students one place to book any lecturer free of licence costs

Conceptual and Technical Direction

Two ideas guide the design From the Technology Acceptance Model every screen must make booking quicker than sending a message or students and lecturers will not switch (Davis, 1989) From the healthcare studies confirmations and reminders are what reduce missed meetings so they are core features rather than extras (Guy, 2012; Zhao, 2017) Lecturers keep full control of their own slots which answers the loss of control concern raised by Zhao (2017)

Technically CampusBook follows a simple three-tier design The React front end runs in the browser the Express API handles logins slots and bookings and the MySQL database stores the data with rules that stop a slot being booked twice scheduled task sends reminder emails before each consultation

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Appendix A: Project Charter and Work Plan

Project Overview

CampusBook is a web based consultation booking system for CTU Training Solutions Students currently arrange consultations through email Teams WhatsApp and walk-ins which leads to missed and unplanned meetings The project will build and evaluate a system that gives students and lecturers one place to manage consultations

Project	CampusBook
Client	CTU Training Solutions students and lecturers
Supervisor	Jabulile Mashibini
Group Lead	Willem Gerber
Method	Scrum with one-week sprints
Duration	16 October to 29 October 2026

Aim and Objectives

The aim is to design develop and evaluate CampusBook so that fewer consultations at CTU are missed or left unplanned the five objectives are to analyse current practice compare existing systems build the six must have features test the system and evaluate it with real users

Scope and Deliverables

The project covers one CTU campus student and lecturer roles and the six must have features It excludes SMS video calls native apps and links to CTU live systems The deliverables are:

- A working web application with source code on GitHub
- A dissertation covering Chapters 1 to 6 with testing evidence
- A live demonstration and presentation on 29 October 2026

Team and Roles

Member	Role	Responsible for
Willem Gerber	Group Lead and Scrum Master	Stand ups GitHub board booking logic compiling the dissertation
Oscar Khumalo	Back-end developer	Express API login and roles back-end write-up
Thomas du Plessis	Front-end developer	React pages mobile layout wireframes and screenshots
Lebohang Tlaitlai	Testing and research lead	Student survey literature review test cases
Gerhardus van den Berg	Database developer	MySQL schema and ERD email reminders

Milestones

Sprint	Main tasks	Milestone
Sprint 0 16 to 19 Oct	Chapters 1 to 3 requirements system design charter ethics form GitHub set-up and first commit	Week 2 and 3 PDFs due Monday 19 Oct 12pm
Sprint 1 20 to 23 Oct	Database login slots and booking reminders history mobile layout student survey testing	Working app and Chapters 4 and 5 due Friday 23 Oct 12pm
Sprint 2 24 to 29 Oct	User testing bug fixes full dissertation Turnitin check presentation practice	Final submission and live demo Thursday 29 Oct

Risks and Assumptions

Risk	Level	How we handle it
Not enough time to build every feature	High	Build the must-haves first and commit daily
Work falls on one or two people	High	Clear task owners checked at every stand up and on GitHub
Some members are new to React or Express	Medium	Pair programming and a shared starter project
Too few survey or test participants	Medium	Share the survey early in class and on Teams

Appendix B: Ethical Clearance Confirmation

Project Title	CampusBook: a lecturer consultation booking system for CTU Training Solutions
Group Members	Willem Gerber Oscar Khumalo Thomas du Plessis Lebohang Tlaitlai Gerhardus van den Berg
Module	Software Engineering 2
Supervisor	Jabulile Mashibini
Week / Stage	Week 2
Submission Date	Monday 19 October 2026

Data and Participants

Activity	Involved	Details
Personal data	Yes	Names emails and bookings with only made-up data during development
Interviews	No	No formal interviews our lecturer reviewed the proposal and supported the idea
Survey	Yes	Anonymous at least 10 CTU students no names or student numbers
User testing	Yes	At least 5 students and 2 lecturers with a short questionnaire
Minors or sensitive topics	No	Adults only questions are only about booking consultations

Ethical Considerations

- Consent: every participant gets a short information sheet and takes part voluntarily and can stop at any time User testing participants sign a consent form and survey participants tick a consent box before starting
- Confidentiality: survey answers are anonymous testing notes use codes such as S1 instead of names and results are only reported as totals or anonymous quotes
- Secure handling: research data is kept in a private password-protected folder and deleted once the project is marked
- Data minimisation: the system only collects what it needs hashes passwords and shows users only their own bookings in line with POPIA

Appendix C: Literature Review Matrix

Source	Focus and method	Key finding	Use and limitation
Griffin (2014)	Office hour use student survey	Most barriers are outside the lecturer control	Shows the problem but only one US university
Weimer (2015)	Office hour attendance survey summary	66% of students never attended	Shows the scale but not peer reviewed
Zhao (2017)	Online medical booking systematic review	Fewer no-shows shorter waits less staff work	Supports online booking but healthcare only
Guy (2012)	SMS reminders meta-analysis	Reminders increase attendance	Supports reminders but tested SMS not email
Davis (1989)	Technology acceptance model	Usefulness and ease of use drive adoption	Guides evaluation but an older workplace study
Nielsen (1994)	Usability heuristics	Ten rules for usable interfaces	Guides design but based on expert opinion
Microsoft (n.d.)	Microsoft Bookings product documentation	Strong Outlook and Teams links business focus	Comparison system vendor source
Calendly (n.d.)	Calendly pricing page	Free plan limited reminders paid	Comparison system vendor source
Moodle (2017)	Moodle Scheduler documentation	Built for teaching but tied to one course	Comparison system page may be dated
Stack Overflow (2025)	Developer survey	React Node.js and MySQL are widely used	Supports the stack but a self-selected sample

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